Projects List for DBMS

B. Tech CSE 3rd Year (Sections E and F only)

DBMS Project Titles:

1. Healthcare and Clinical Management
2. Education and Academic Management
3. Library and Knowledge Resource Management
4. Banking and Financial Services
5. Insurance and Claims Management
6. Retail and Inventory Management
7. E-Commerce and Order Management
8. Restaurant and Food Service Management
9. Hotel and Hospitality Management
10. Travel and Tourism Management
11. Transportation and Fleet Management
12. Railway and Public Transit Management
13. Vehicle Rental and Service Management
14. Real Estate and Property Management
15. Human Resource and Employee Management
16. Recruitment and Workforce Management
17. Manufacturing and Production Management
18. Supply Chain and Procurement Management
19. Warehouse and Logistics Management
20. Hospitality and Event Management
21. Sports and Recreation Management
22. Fitness and Wellness Management
23. Entertainment and Media Management
24. Pharmaceutical and Medicine Management
25. Blood Donation and Medical Resource Management
26. Agriculture and Farm Management
27. Food Supply and Distribution Management
28. Energy and Utility Management

29. Waste Management and Recycling
30. Water Resource and Distribution Management
31. Smart City and Municipal Services
32. Public Service and E-Governance
33. Security and Incident Management
34. Subscription and Membership Management
35. AI/Technology-Enabled Management Systems

General Project Instructions for implementing the project:

DBMS Core Logic and SQL Implementation

1. Identify entities, attributes, relationships and business rules. Develop the ER/EER model and convert it into a relational schema.
2. Identify primary keys, candidate keys and foreign keys. Identify functional dependencies and demonstrate normalization up to 3NF/BCNF, where applicable.
3. Implement the database using DDL and DML SQL commands. Demonstrate joins, nested/correlated queries, sub queries, aggregate functions, GROUP BY, HAVING and etc. Use sufficient and realistic sample data.
4. Create appropriate views and indexes. Implement triggers and/or stored procedures where applicable. Demonstrate transactions and ACID properties and discuss concurrency where relevant. Implement basic database security and role-based access control where applicable.

Front-End Design and Database Interaction: Develop a simple web-based front end connected to the database. Provide interfaces for insert, search, update, delete and report generation. Illustrate the SQL queries executed through the front end and display their results. Provide a button/link to view the ER model/database schema from the front end. Demonstrate the complete flow: Front End → SQL Query → Database → Result. The DBMS implementation and core database logic should remain the primary focus; the front end should support and demonstrate the database functionality.

Artificial Intelligence / Machine Learning Integration: Identify and implement at least one meaningful AI/ML task using project data, including appropriate data preprocessing, model selection, training, testing, and performance evaluation. Integrate the AI/ML component with the database and front end to demonstrate a useful intelligent feature such as prediction, classification, recommendation, forecasting, anomaly detection, or intelligent search.